Jaume Puig

Space Radiobiologist and Engineer

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PROFESSIONAL EXPERIENCE

Oct 2024 to Present

CTO & Co-Founder
Barding Industries
- Architecting the technology platform: engineered biology that recovers critical minerals, including rare earth elements, from ores and waste streams.

Jun 2024 to Present

Communications Coordinator & Member, Microbes & Hardware AWGs
NASA Open Science Data Repository
- Contributing to open-science space microbiology through omics data reanalysis, pipeline optimization, and methods standardization.
- Co-authoring the SPLASH Database paper alongside AWG principal investigators.
- Managing internal communications and inter-subgroup coordination across the AWG.

Feb 2021 to Present

Judge, Mentor & Community Member
iGEM Foundation
- Executive Judging Committee Member (2026 to Present): guiding the evolution and quality of judging across the competition, from judging philosophy, standards, and rubrics to training incoming judges and ensuring fairness and consistency.
- Space Network Member (2024 to Present): advising teams on space exploration and astrobiology applications.
- Judge (2024 to Present): evaluating international synthetic biology projects.
- Mentor (2022 to Present): technical guidance across experimental design and modeling.
- Ambassador for Europe (2022 to 2024): promoted synthetic biology across Europe.

May 2024 to Present

STEAM Educator
AulaSPUTNIK
- Designing and delivering science workshops for children, and managing the organization's website.

Jun 2024 to Jan 2025

Research Fellow, Crew 2: The Illuminator's Class
BioAstra
- Analyzed pharmacogenomic and nutritional genomics datasets from spaceflight missions to advance commercial astronaut health research.
- Contributed to protocol development for the Space Omics and Medical Atlas (SOMA) framework.

Apr 2023 to Jan 2024

Space Biotechnology Technician
Yuri GmbH
- Conducted microbial culture experiments and multi-omics analysis for ISS payload projects using automated modular bioreactors.
- Managed laboratory facilities, hardware testing, and data interpretation across multiple concurrent payload programs.

ACADEMIC BACKGROUND

2025 to Present

Ph.D. in Biomolecular Sciences
Laurentian University / SNOLAB
- Major: Radiation Biology.
- McLaughlin Graduate Scholarship recipient.
- Teaching Assistant.

2022 to 2023

M.Sc. in Multidisciplinary Research in Exp. Sciences
BIST / UPF

- Major: Low-Background Radiation Omics. Minor: Ancestral Gene Resurrection.
- JAE Intro ICU Grant (CSIC).
- Lead organizer of ScienceHack 2023 (BIST-MMRES), a science hackathon with 360+ participants.

2016 to 2022

B.Eng. in Biomedical Engineering
Universitat Pompeu Fabra (UPF)

- Major: LEO Radiation Biotolerance.
- iGEM 2020: Gold Medal, Best New Basic Part, Best Wiki. Nominated for Best Therapeutics Project, Best Poster, and Best Presentation.

PUBLICATIONS

2025

STREAMS guidelines: standards for technical reporting in environmental and host-associated microbiome studies
Kelliher, Mirzayi, Bordenstein, … STREAMS Consortium, et al. Nature Microbiology 10, 3059–3068.

2025

Feeding the cosmos: tackling personalized space nutrition and the leaky gut challenge
Barbero Barcenilla, Rivero, Lynch, … **Puig**, et al. npj Microgravity 11, 45.

2024

Intein-mediated thyroid hormone biosensors: towards controlled delivery of hormone therapy
Martí-Baena, Pascuet-Fontanet, Berjaga-Buisan, Caravaca-Rodríguez, **Puig-Costa-Jussà**, et al. Frontiers in Systems Biology 4:1270071.

2021

DNA damage protection for enhanced bacterial survival under simulated low Earth orbit environmental conditions in Escherichia coli
Puig, Knödlseder, Quera, Algara, Güell. Frontiers in Microbiology 12:789668.

2024

Space Omics and Medical Atlas (SOMA) protocol compilation
Contributor. Open protocols for spaceflight omics and medical research.

2020

Hormonic: intein-mediated thyroid hormone biosensors
iGEM 2020, Team UPF_Barcelona wiki. Gold Medal, Best New Basic Part, Best Wiki.

PRESENTATIONS

Oct 2026

Global consistency of biological responses to sub-terrestrial low-background radiation environments
Radiation Research Society (RRS) 72nd Annual Meeting, Waikoloa, Hawaii, USA.

Oct 2024

Spaceflight effects on human skin microbiome
International Astronautical Congress (IAC) 2024, Milan, Italy.

Oct 2023

DNA repair and genetic radiation biotolerance
International Astronautical Congress (IAC) 2023, Baku, Azerbaijan.

Sep 2022

DNA damage protection for enhanced bacterial survival under simulated low Earth orbit conditions
Europlanet Science Congress 2022, Granada, Spain.

May 2022

DNA damage protection for enhanced bacterial survival under simulated low Earth orbit conditions
AGU / NASA Astrobiology Science Conference (AbSciCon), Atlanta, USA.

AWARDS & HONORS

2025

McLaughlin Graduate Scholarship Laurentian University.

2022 to 2023

JAE Intro ICU Grant Spanish National Research Council (CSIC), awarded during the M.Sc.

2022

NASA Space Apps Challenge Finalist Presented at the NASA TOPS Forum (virtual, April 2023): “Measuring Open Science”.

2020 **iGEM Gold Medal** International Genetically Engineered Machine competition, Team UPF_Barcelona.

2020 **Best New Basic Part** iGEM 2020, Team UPF_Barcelona.

2020 **Best Wiki** iGEM 2020, Team UPF_Barcelona.

2020 **Best Therapeutics Project, Best Poster, Best Presentation** iGEM 2020, Team UPF_Barcelona.